

Java OOP Case Study - Smart Banking Management System (Student Version)

Scenario

ABC Bank is developing a Java-based Banking Management System. The application should manage customers, accounts, loans, transactions, and employees while demonstrating Java OOP concepts such as encapsulation, inheritance, polymorphism, abstraction, interfaces, collections, and exception handling.

Business Requirements

- Register customers
- Open Savings, Current and Fixed Deposit accounts
- Deposit, withdraw and transfer money
- Apply for loans
- Maintain transaction history
- Generate reports

Sample Input

Customer ID: C101
Customer Name: Rahul Sharma
Account Number: 100101
Account Type: Savings
Opening Balance: 50000
Deposit Amount: 5000
Withdrawal Amount: 3000

Programming Tasks (Questions Only)

1. Identify the classes required for the system.
2. Design the class hierarchy using inheritance.
3. Identify attributes and methods for each class.
4. Implement encapsulation using private data members.
5. Create constructors for each class.
6. Override withdraw() for different account types.
7. Overload deposit() with multiple parameter lists.
8. Create an abstract Account class.
9. Implement a BankOperations interface.
10. Demonstrate runtime polymorphism.
11. Store Account objects using ArrayList.
12. Search an account by account number.
13. Find the account with the highest balance.
14. Calculate yearly interest for each account type.
15. Transfer money between two accounts.
16. Maintain transaction history.
17. Handle invalid deposits and insufficient balance using exceptions.
18. Display account details using toString().
19. Draw the UML class diagram.
20. Suggest three enhancements to the application.

Expected Output

Customer Registered Successfully
Account Created Successfully
Deposit Successful
Updated Balance: 52000
Interest Calculated Successfully
Transaction Completed Successfully

Java Concepts Covered

Classes & Objects, Constructors, Encapsulation, Inheritance, Method Overloading, Method Overriding, Runtime Polymorphism, Abstraction, Interfaces, Collections, Exception Handling, Static Members, final Keyword, toString(), equals().